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What is DevOps?

DevOps is a culture and set of practices that combines **software development (Dev)** and **IT operations (Ops)** to improve collaboration and automate software delivery.

Key Goals

- **Faster Software Delivery**
Automates development and deployment processes.
- **Better Collaboration**
Developers and operations teams work together instead of separately.
- **Continuous Integration and Deployment**
Code changes are integrated, tested, and deployed automatically.
- **Improved Reliability**
Automation and monitoring reduce errors and downtime.

Dev vs Ops Gap

In traditional IT environments, **Development and Operations teams worked separately**, which created communication and workflow problems.

Key Issues

- **Different Goals**
Developers focus on releasing new features quickly, while operations focus on system stability.
- **Communication Gap**
Limited collaboration leads to misunderstandings and delays.
- **Slow Deployment**
Manual processes slow down software releases.
- **Blame Culture**
When systems fail, teams often blame each other.

DevOps Solution

DevOps removes this gap by **encouraging collaboration, automation, and shared responsibility**.

Traditional vs Agile

Traditional Model (Waterfall)

- **Sequential Process:** Development phases occur one after another.
- **Late Testing:** Testing happens mostly after development is completed.
- **Slow Delivery:** Large releases take months or years.
- **Less Flexibility:** Changes are difficult once development has started.

Agile Model

- **Iterative Development:** Software is built in small, incremental cycles.
- **Continuous Feedback:** Regular testing and customer feedback during development.
- **Faster Releases:** Features are delivered frequently in small updates.
- **Adaptable to Changes:** Requirements can evolve during development.

Software Development Lifecycle (SDLC)

The **Software Development Lifecycle (SDLC)** is a structured process used to design, develop, test, and deploy software efficiently and with high quality.

Stages of SDLC

- **Planning:** Define project goals, scope, feasibility, and required resources.
- **Requirement Analysis:** Gather and document user and business requirements.
- **Design:** Create system architecture, database design, and application structure.
- **Development (Coding):** Developers write the actual code based on design specifications.
- **Testing:** Identify bugs and ensure the software meets requirements.
- **Deployment:** Release the application to production for users.
- **Maintenance:** Fix issues, improve performance, and add new features.

DevOps Lifecycle (Definition)

The **DevOps Lifecycle** is a continuous and automated process that integrates software development and IT operations to deliver applications faster, more reliably, and with higher quality. It emphasizes collaboration, automation, continuous feedback, and rapid delivery of software updates.

Unlike traditional development models where software is built and released in large cycles, the DevOps lifecycle promotes **continuous integration, continuous testing, continuous deployment, and continuous monitoring**.

Stages of DevOps Lifecycle

The DevOps lifecycle consists of several interconnected stages that enable continuous software delivery.

Plan: Teams define project goals, gather requirements, and plan development tasks.

Develop: Developers write and manage application code using version control systems.

Build: Source code is compiled and converted into executable software packages.

Test: Automated testing is performed to identify bugs and ensure software quality.

Release: The tested build is prepared and approved for deployment.

Deploy: The application is deployed to production or staging environments.

Operate: The deployed application is managed and maintained to ensure smooth operation.

Monitor: System performance and user experience are monitored to detect issues and improve future releases.